Simulation of Nanosystems for Energy Conversion
Department of Electrical and Computer Engineering
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# Gaussian Process Regression as a surrogate model for estimating residential building energy loads

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**Abstract** — Gaussian Process Regression (GPR) is evaluated as a surrogate model for predicting heating and cooling loads of residential buildings from eight geometric design parameters, using 768 simulated configurations from the Energy Efficiency dataset. Under 5-fold cross-validation, a GPR model with an anisotropic Matern kernel reaches a root mean squared error of 0.29 for the heating load and 0.39 for the cooling load, improving on ordinary linear regression by a factor of 9.8 and 8.2 respectively. Two ablation studies identify Automatic Relevance Determination as the decisive modelling choice, while the kernel family matters for both predictions. The predictive intervals are conservative at the 68% level and close to nominal at the 95% level, making the surrogate useful in design contexts where the reliability of a prediction matters as much as its accuracy.

## 1 Introduction

Numerical simulations are widely used in engineering to study physical systems and evaluate different design choices. However, large-scale numerical simulations demand considerable computational resources, particularly when they must be evaluated repeatedly for design, optimization, or uncertainty analysis [1]. In building-energy analysis, advanced simulation software can also be time-consuming to use and requires expertise in the selected program [2].

Surrogate models offer an alternative by using machine learning to provide fast approximations of the target variables at new design points. Once trained on data generated by the numerical simulation, a surrogate model can predict new configurations much faster than rerunning the original simulation [3]. Python provides a suitable environment for this workflow through several libraries. `pandas` and `matplotlib` can be used to explore and visualize the data, while `scikit-learn` provides simple and efficient tools for predictive data analysis, including regression, preprocessing, cross-validation, and model evaluation.

Residential building-energy analysis is used in this project as an application of surrogate modeling. Heating, ventilation, and air-conditioning systems account for a large proportion of building energy consumption [4], making accurate estimates of heating load (HL) and cooling load (CL) important for energy-efficient building design and equipment sizing [2].

This project investigates Gaussian Process Regression (GPR) as a surrogate model for this dataset. Ordinary linear regression and Kernel Ridge Regression are used as reference models, while different GPR kernel configurations and their predictive uncertainty are also evaluated. The project therefore addresses the following research question: *How accurately and reliably can Gaussian Process Regression predict residential heating and cooling loads as a surrogate model compared with simpler regression approaches?*

## 2 Data Description

The Energy Efficiency dataset of Tsanas and Xifara [2] contains 768 residential building configurations simulated with Ecotect. The eight design variables of each configuration are: relative compactness ($X_1$), surface area ($X_2$), wall area ($X_3$), roof area ($X_4$), overall height ($X_5$), orientation ($X_6$), glazing area ($X_7$), and glazing-area distribution ($X_8$). The prediction targets are heating load ($Y_1$) and cooling load ($Y_2$).

The configurations were generated systematically from twelve building forms of equal volume ($771.75\,\mathrm{m}^3$), while orientation and glazing parameters were varied. Climate, materials, occupancy, and operating conditions were kept fixed [2]. For this reason, the dataset does not represent a random sample of real-world buildings, but rather a systematically created simulation design.

The distributions of the input variables are shown in Figure 1. Several inputs take only a small number of values. For example, overall height ($X_5$) has two possible values, while orientation ($X_6$) and glazing area ($X_7$) each have four. Orientation ($X_6$) and glazing-area distribution ($X_8$) represent categorical design choices.

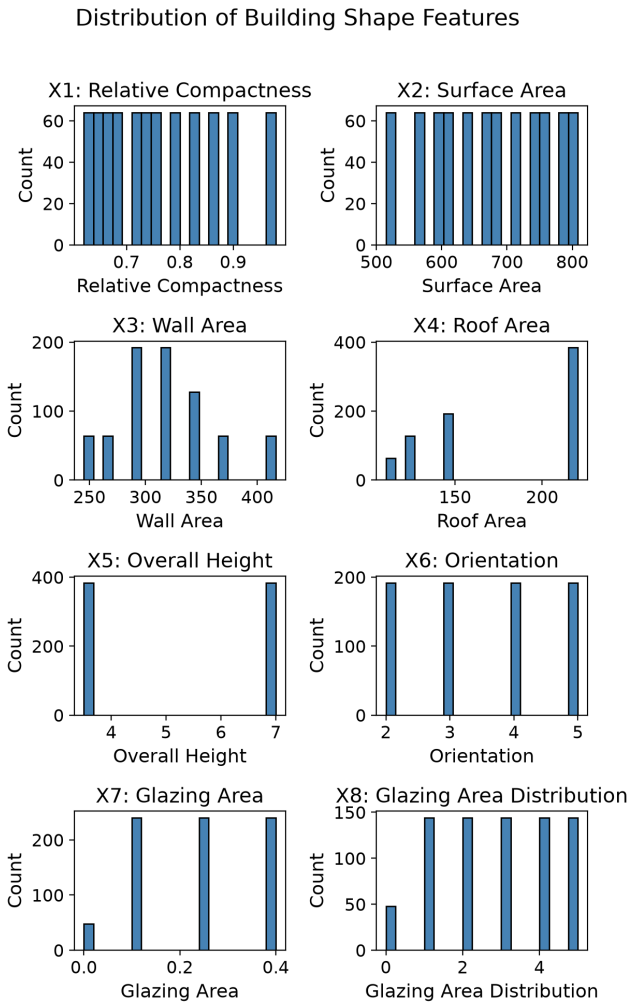



**Figure 1** Distributions of the eight building design parameters.

Another observation is that the targets, HL and CL, are not normally distributed, as shown in Figure A1 in the appendix.

Figure A2 in the appendix shows the Spearman correlations between the input variables and the two targets. Overall height ($X_5$) and roof area ($X_4$) show very strong monotonic relationships with both heating and cooling load, whereas orientation ($X_6$) shows no monotonic association. Strong dependencies also occur between some of the geometric inputs. Very strong negative correlation is present between relative compactness ($X_1$) and surface area ($X_2$), and also between roof area ($X_4$) and overall height ($X_5$). These dependencies partly result from the fixed-volume building design [2].

# 3 Methods

Heating load ($Y_1$) and cooling load ($Y_2$) were modeled separately using the same eight building parameters as inputs. Three regression methods were implemented using `scikit-learn`: Linear Regression as a simple baseline, Kernel Ridge Regression (KRR) as a nonlinear kernel-based reference, and Gaussian Process Regression (GPR) as the main surrogate model.

## 3.1 Preprocessing and Evaluation

A first challenge was that the input variables have different types and scales. Six variables ($X_1, X_2, X_3, X_4, X_5, X_7$) were treated as numerical and standardized to zero mean and unit variance. Orientation ($X_6$) and glazing-area distribution ($X_8$) were treated as categorical variables and one-hot encoded.

Standardization and one-hot encoding were fitted only on the training data of each cross-validation fold. The fitted preprocessor was then applied to the corresponding validation data to avoid information leakage.

Model performance was evaluated using shuffled five-fold cross-validation with a fixed random seed. The same folds were used for all models to make their results directly comparable. Prediction accuracy was measured using root mean squared error (RMSE) and the coefficient of determination ($R^2$), reported as mean ± standard deviation across the five folds.

Linear Regression does not require hyperparameter tuning and was used as the main baseline. KRR was implemented with an RBF kernel. A challenge was selecting its regularization parameter $\alpha$ and kernel parameter $\gamma$ without using information from the outer validation fold. Because of that, we selected them using shuffled three-fold `GridSearchCV` within each outer training fold.

## 3.2 Gaussian Process Regression

GPR was selected as the main surrogate model because it can represent nonlinear relationships while also providing predictive uncertainty. Instead of assuming one fixed function, a Gaussian process defines a distribution over possible functions [5]. Its kernel describes how strongly function values at different input points are related [5].

An important modelling question was which level of smoothness should be assumed for the relationship between the building parameters and the loads. So, we tested three kernel types separately for heating and cooling load: the RBF kernel and Matérn kernels with

$\nu = 1.5$ and $\nu = 2.5$. The RBF kernel assumes a very smooth function, while the Matérn kernels allow less smooth variation through the parameter $\nu$ [5]. The kernels were compared using cross-validated RMSE, and for each target the best-performing kernel was selected.

Each kernel also contained a constant term controlling the overall signal amplitude and a white-noise term. The signal amplitude, noise level, and kernel length scales were optimized on the training data by maximizing the log marginal likelihood. This hyperparameter optimization is built into `scikit-learn`'s Gaussian Process Regression implementation. Multiple optimizer restarts were used to reduce sensitivity to local optima.

Another modelling challenge was that the different input dimensions may influence the target on different characteristic scales. Automatic Relevance Determination (ARD) was therefore implemented by assigning an individual length scale to each preprocessed input dimension. Smaller length scales allow the fitted function to change more rapidly along that dimension, whereas larger length scales represent slower variation [5]. The eight original variables resulted in 16 model inputs after one-hot encoding $X_6$ and $X_8$, and therefore in 16 length scales. To evaluate whether this additional flexibility was important, the selected GPR models were also compared with isotropic versions using one shared length scale.

### 3.3 Predictive Uncertainty

One advantage of GPR over the other regression models is that it provides both a predicted value and an estimate of predictive uncertainty. To evaluate this uncertainty without assessing a model on samples it had already seen, out-of-fold predictions were used. Each sample was therefore predicted by a model that had not been trained on that sample.

For each prediction, GPR returned a predictive mean $\hat{y}$ and a standard deviation $\sigma$. We measured how often the true value fell within the intervals $\hat{y} \pm \sigma$ and $\hat{y} \pm 1.96\sigma$. For approximately Gaussian predictive distributions, these intervals correspond to nominal coverage levels of about 68% and 95%, respectively.

## 4 Results and Discussion

### 4.1 Model Comparison

Table 1 summarises the predictive accuracy of the three model families. Ordinary linear regression already explains a large share of the variance, consistent with the strong linear correlations between the geometric features and both targets reported in Section 2. Its RMSE nevertheless remains close to three load units for both targets, indicating that a substantial part of the relationship is not linear.

**Table 1** Cross-validated performance of all models (mean ± std over 5 folds). GP uses Matern($\nu = 1.5$) for Y1 and Matern($\nu = 2.5$) for Y2. Continuous features are standardized and categorical features (X6, X8) are one-hot encoded within each fold. Standard deviations of $R^2$ are below 0.009 throughout and are omitted.

| Model | Target | RMSE | $R^2$ |
|---|---|---|---|
| Linear Regression | Y1 | 2.827 ± 0.146 | 0.920 |
| | Y2 | 3.194 ± 0.131 | 0.886 |
| KRR (RBF) | Y1 | 0.614 ± 0.034 | 0.996 |
| | Y2 | 1.490 ± 0.152 | 0.975 |
| GP (Matern, ARD) | Y1 | 0.288 ± 0.117 | 0.999 |
| | Y2 | 0.392 ± 0.026 | 0.998 |

Kernel Ridge Regression (KRR) improves substantially on the linear baseline for both targets once the inner cross-validation folds are correctly shuffled and the categorical inputs are one-hot encoded, reducing RMSE by a factor of 4.6 for the heating load and 2.1 for the cooling load. During tuning, $\alpha$ repeatedly settled at the lower edge of the search grid; extending the grid further caused the RBF kernel matrix to become numerically degenerate, so this was documented as a limitation rather than resolved further.

The Gaussian Process (GP) with an anisotropic Matern kernel clearly outperforms both baselines, reducing RMSE by a factor of 9.8 (heating load) and 8.2 (cooling load) relative to Linear Regression, and by a factor of 2.1 and 3.8 relative to KRR. For the cooling load it is also the most stable model across folds (standard deviation 0.026, versus 0.131 for Linear Regression and 0.152 for KRR); for the heating load, KRR is marginally more stable (0.034) than the GP (0.117), despite its substantially higher error. A graphical comparison is provided in Figure A3 in the appendix.

For external context, Tsanas and Xifara reported a random-forest MAE of 0.51 for the heating load and 1.42 for the cooling load under their own validation protocol [2]. The values are not directly comparable, since RMSE and MAE differ and the cross-validation schemes are not identical, but both studies confirm that nonlinear models clearly outperform linear baselines on this dataset.

## 4.2 Kernel Choice and the Role of ARD

Two ablation studies were carried out to identify which modelling decision drives the performance of the GP.

The first compares kernel families. After correcting the categorical encoding, kernel choice matters for both targets: for the heating load, Matern($\nu = 1.5$) reaches an RMSE of 0.288, ahead of Matern($\nu = 2.5$) (0.329) and RBF (0.463); for the cooling load, Matern($\nu = 2.5$) is best (0.392), ahead of RBF (0.437) and Matern($\nu = 1.5$) (0.491). Notably, the most accurate kernel is not the most stable one: RBF has the lowest fold-to-fold variability for both targets (standard deviation 0.040 for Y1, 0.017 for Y2), even though it is the least accurate kernel for the heating load and only the second-most accurate for the cooling load. The results indicate that a finite-differentiability (Matérn) kernel provides the best fit for both targets, though the optimal smoothness parameter differs between heating and cooling load.

The second ablation isolates the effect of Automatic Relevance Determination (ARD), that is, one length scale per feature (or, after one-hot encoding, per category) instead of a single shared one. Removing ARD increases the RMSE from 0.288 to 0.588 for the heating load, roughly doubling the error, and from 0.392 to 1.390 for the cooling load, more than a threefold increase. ARD is therefore the single most influential modelling choice in this study. Rather than one length scale per raw feature, ARD now assigns an individual length scale to each category; for the cooling load, one glazing-area-distribution category shows a markedly smaller length scale than the others, indicating that its predictive relevance is highly category-specific rather than smoothly varying across the original ordinal code. Given the strong collinearity among the geometric features (Section 2), these patterns reflect predictive sensitivity within the observed design rather than an isolated physical mechanism.

Notably, the isotropic GP still slightly outperforms KRR for both targets (0.588 versus 0.614 for the heating load, 1.390 versus 1.490 for the cooling load), but the margin is much narrower than before the preprocessing fixes. This indicates that most of the GP's advantage over KRR is attributable to ARD, with only a small residual contribution from the marginal-likelihood optimisation of the remaining hyperparameters and the explicit noise term.

**Table 2** Observed coverage of the GP predictive intervals, computed from out-of-fold predictions.

| Target | Nominal 68% | Nominal 95% |
|---|---|---|
| Y1 Heating Load | 0.797 | 0.938 |
| Y2 Cooling Load | 0.786 | 0.940 |

## 4.3 Predictive Uncertainty

Beyond point accuracy, the GP provides a predictive standard deviation. To assess whether this uncertainty is trustworthy, the empirical coverage of the predicted intervals was computed from out-of-fold predictions (Table 2).

Figure 2 shows this pattern for both targets. At the 68% nominal level, observed coverage (0.797 for Y1, 0.786 for Y2) exceeds the nominal value, meaning the intervals are wider than necessary and the model is conservative. At the 95% nominal level, observed coverage (0.938 for Y1, 0.940 for Y2) is close to, but still slightly below, the nominal value, so the intervals are marginally too narrow in the tails; matching the nominal level exactly would require a slightly larger multiplier than 1.96. The overall deviation from ideal calibration is small, and the uncertainty estimates can be regarded as broadly trustworthy for this dataset. In Figure A4 in the appendix, the predictive standard deviation widens in sparsely populated regions of the target range, that is, the model increases its stated uncertainty where fewer similar training points are available in target space. This behaviour was not evaluated for extrapolation in input space and should not be read as a general guarantee of out-of-domain detection.

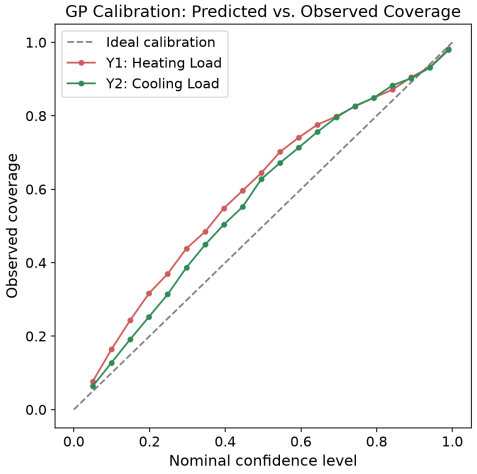



**Figure 2** Calibration of the GP predictive intervals. Observed coverage is plotted against the nominal confidence level; the dashed line marks ideal calibration.

## 5 Conclusion and Limitations

Gaussian Process Regression is an effective surrogate model for the residential building energy dataset of Tsanas and Xifara [2]. Using an anisotropic Matern kernel with categorical inputs one-hot encoded, it predicts the heating load with an RMSE of 0.29 and the cooling load with an RMSE of 0.39, improving on ordinary linear regression by a factor of 9.8 and 8.2 respectively, and on a tuned Kernel Ridge model by a factor of 2.1 and 3.8. Two ablation studies identify ARD as the decisive modelling choice for both targets; kernel family also matters, though the optimal smoothness parameter differs between heating and cooling load. The predictive intervals are conservative at the 68% level and close to nominal at the 95% level.

Several limitations should be stated. First, the 768 samples form a factorial design of twelve building shapes combined with four orientations and several glazing configurations rather than a random sample; every validation point therefore has close neighbours in the training set, and the reported errors describe interpolation within a regular grid rather than generalisation to arbitrary buildings.

Second, all samples share the same volume of approximately $772\,\mathrm{m}^3$ and a single climate and material assumption; predictions outside this envelope, for example larger non-residential structures, are unsupported by the data and were not tested here.

Third, the targets originate from a simulation, so the models approximate the Ecotect model rather than physical reality. The geometric features are also strongly intercorrelated ($|r| > 0.85$ between several of them), so the learned length scales indicate predictive relevance within this design rather than a causal or purely physical relationship. Finally, exact GP inference scales cubically with the number of training samples, which limits direct applicability to substantially larger datasets [5].

## 6 AI-Use Statement

Generative AI tools were used in the preparation of this project. Anthropic's Claude was used through its chat interface and through the Claude Code extension for Visual Studio Code.

For the implementation, Claude produced initial scaffolding for data retrieval and a first GPR script, and generated the analysis scripts for the exploratory analysis, kernel model selection, baseline comparison, ARD ablation and uncertainty evaluation from our written specifications. We defined the experimental design, reviewed the generated code, and ran and validated all reported results ourselves.

For the text, Claude produced first drafts of Sections 4 and 5, which we revised, corrected, and shortened. For Sections 1, 2, and 3, ChatGPT was used to turn our text into a more continuous and coherent text. So ChatGPT was used to enhance writing and comprehension. ChatGPT and Claude were also used to clarify methodological concepts such as GPR kernels, Automatic Relevance Determination, covariance functions, and interval calibration.

All modelling decisions, the interpretation of the results and the selection of literature are our own. Every reference was verified against the original publication. We also identified and corrected factual errors in AI-generated text, including an overstated comparison between the isotropic Gaussian Process and Kernel Ridge Regression. The authors take full responsibility for the correctness of the content presented here.

## 7 References

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[5] C. E. Rasmussen and C. K. I. Williams, Gaussian Processes for Machine Learning. Cambridge, MA, USA: MIT Press, 2006.

# 8 Appendix

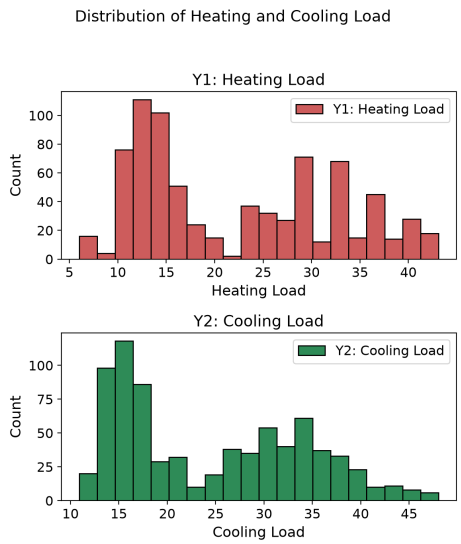



**Figure A1** Distributions of heating load ($Y_1$) and cooling load ($Y_2$).

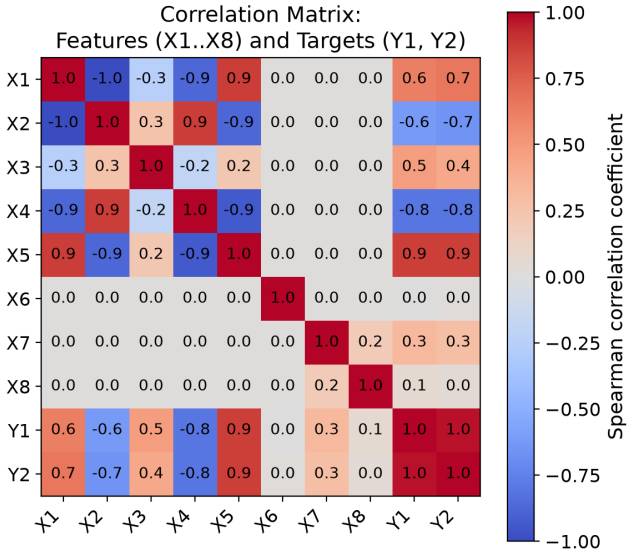



**Figure A2** Spearman correlation matrix of the eight input variables and the two regression targets.

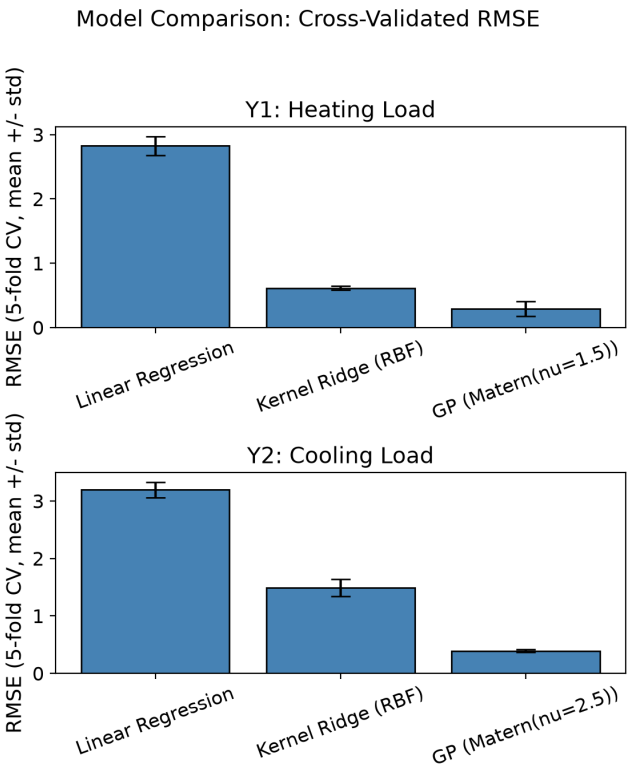



**Figure A3** Cross-validated RMSE of the three model families for heating load (top) and cooling load (bottom). Error bars show the standard deviation across the five folds. The same values are given numerically in Table 1.

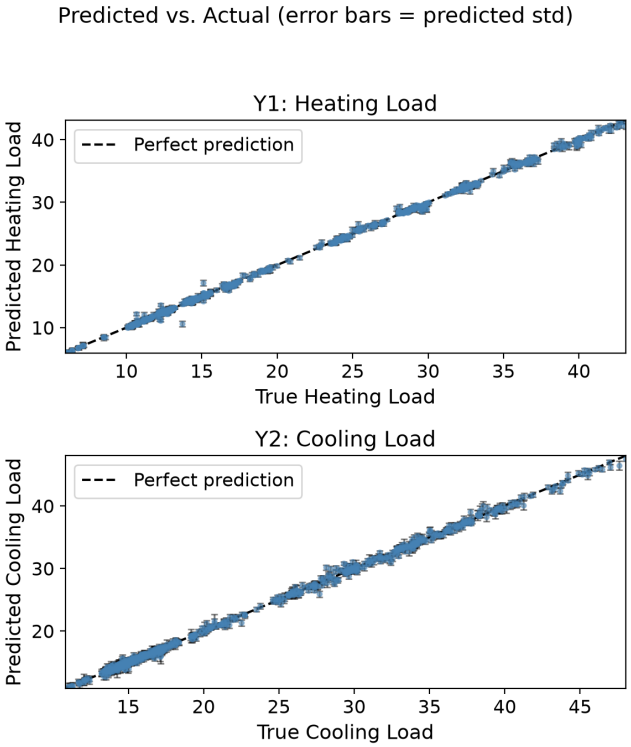



**Figure A4** Out-of-fold predictions against true values, with error bars given by the predicted standard deviation.